

ImmersiveAutochrome: Enhanced Digitized Mono-Autochrome Viewing through Deep Learning and Virtual Reality

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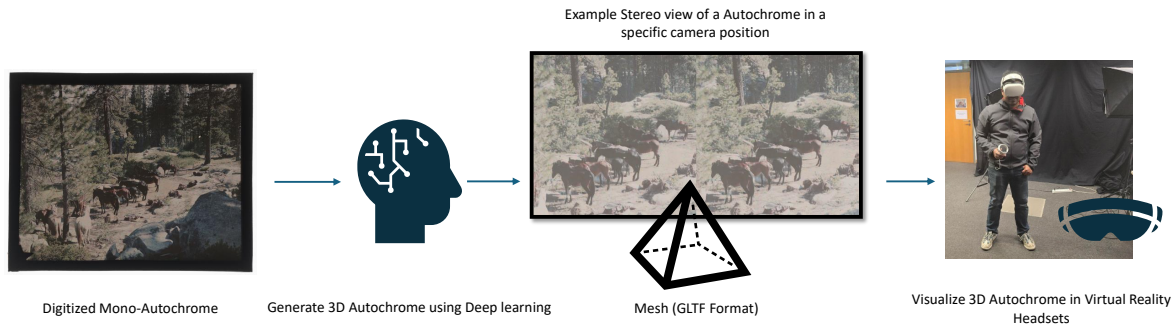


Figure 1: The pipeline utilizes deep learning techniques to transform a digitized mono-autochrome into an immersive 3D autochrome (layered depth image with corresponding mesh), enabling visualization through a Virtual Reality (VR) headset.

ABSTRACT

Resulting from the earliest impactful color photography process developed by the Lumière Brothers, autochromes are invaluable historical artifacts with a high sensitivity and susceptibility to degradation over time. We investigate the potential of deep learning to enhance the experience of historical autochromes through 3D photography and Virtual Reality (VR). Our proposed pipeline utilizes single-image depth estimation and depth-aware inpainting to transform digitized mono-autochromes into layered depth images and corresponding 3D meshes, enabling immersive 3D visualization via a WebXR application. Our user study shows the effectiveness of VR-based experience of autochromes compared to traditional visualization methods, revealing that participants found the VR experience as pleasant as analog exploration despite the latter's authenticity. Thereby, our findings suggest that 3D VR technology could play a crucial role in digitally preserving and revitalizing these culturally significant artifacts and foster investigations with more samples and more types of user interfaces to mature these insights.

Index Terms: 3D photography, single-image-based depth estimation, inpainting, virtual reality

1 INTRODUCTION

Color photography has become a vital aspect of contemporary society. The Lumière brothers' invention of the autochromes [12] is an early method of color photography from the beginning of the 20th century and served as an essential step in the development of

color photography. Launched commercially in 1907, autochromes are considered the first successful color photography process and maintained their dominance until the introduction of Kodachrome films in the 1930s, which also makes them represent historical artifacts and documents. Autochrome plates, coated with microscopic red, green, and violet/blue potato starch grains, create a full-color image of the original subject by allowing light to pass through these color filters to the photographic emulsion, which is then processed to produce a positive transparency. Autochromes were typically viewed privately by merely holding them against a light source. However, for a more enhanced viewing experience, they were typically placed on specially designed stands known as diascopes, which contained a mirror. This setup not only brightened the image but also facilitated simultaneous viewing by multiple people. For larger public exhibitions, magic lanterns were employed to project the autochromes. The glass plates used in autochrome photographs are highly sensitive, making damage-free conservation challenging, and various types of defects have developed in these collections over time. Hence, digitized copies of these autochromes are commonly shown to the public compared to the analog autochromes, as the latter are susceptible to degradation. The underlying challenge is to preserve the original look of the autochrome in the best possible way and to maintain the associated cultural and historical context.

3D photography, which involves capturing the world through a camera and employing image-based rendering techniques for novel view synthesis, offers a significantly more immersive experience than traditional 2D photography, approaching lifelike realism in Virtual Reality (VR) and providing enhanced depth perception on flat displays through parallax. In this paper, we exploit deep learning technologies to convert a digitized mono-autochrome to a 3D photo [11, 24] and then visualize them using Virtual Reality headsets [11]. Based on leveraging deep learning technologies like single-image depth estimation and depth-aware inpainting, we enable to view digitized autochromes in modern VR devices which provides a new and immersive perspective to these historical artifacts, enhancing their cultural and historical significance by bringing them to life in a three-dimensional space. Our main contribu-

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Figure 2: Single image depth estimation of different autochrome scenes using Boosting Monocular Depth Estimation [16]

tions include:

- A pipeline that utilizes deep learning technologies, specifically single-image depth estimation and depth-aware inpainting, to convert a mono-autochrome into a layered depth image [22] and corresponding mesh, followed by the creation of a WebXR application for visualization in virtual reality (VR), as shown in Fig. 1.
- A user study conducted to evaluate the comparative effectiveness and impact of viewing autochromes through VR devices versus traditional visualization methods using a back-light board to display analog autochromes, as shown in Fig. 6. The 3D autochromes generated from selected digitized autochromes of the Harold Taylor collection [26] are now available on [GitHub](#) to aid in research progress.

2 RELATED WORKS

2.1 Single-image depth estimation

Monocular depth estimation (MDE) is an advanced task that involves interpreting monocular depth cues, including occlusions, perspective, and the relative sizes of objects. Initial studies [21, 9, 15] mainly depended on handcrafted features and conventional computer vision methods, which were restricted by their reliance on explicit depth cues and difficulties in addressing complex scenes with occlusions and areas lacking texture. Deep learning approaches have transformed monocular depth estimation by efficiently learning depth representations from carefully annotated datasets. In particular, recent deep neural network approaches [5, 34, 28, 31, 23] have been demonstrated to directly learn depth-related priors from training data, resulting in remarkable advancements in performance due to the availability of large-scale depth datasets [6, 25]. To facilitate effective multi-dataset joint training, the groundbreaking work in zero-shot depth estimation achieved based on the MiDaS approach [20] employs an affine-invariant loss that disregards the potentially differing depth scales and shifts present in various datasets, thereby providing relative depth information. In our study, we utilize an improved version of the MiDaS model [3] in conjunction with the approach presented in [16] for depth estimation. While recent advancements in zero-shot depth estimation models [2, 33] have emerged, we did not evaluate these models in our experiments. Investigating the compatibility of 2D autochromes with these models will be an avenue for future research.

2.2 Single-image novel view synthesis

Novel view synthesis allows for the generation of images from novel camera viewpoints not considered during the capture pro-

cess by leveraging 2D images and their related camera poses. Recently, both implicit [1, 18] and explicit [10] learning-based scene representations have demonstrated to offer an impressive potential in generating photorealistic novel views and represent the current state-of-the-art novel view synthesis. Nonetheless, these approaches typically require the availability of many dense views as input, which are not always accessible in many situations and also in the case of 2D mono autochromes. Additionally, they prioritize interpolation using multiple views instead of extrapolation. Thus, we focus on approaches designed to handle single inputs. Several works [27, 7, 14, 32] aim to infer the 3D structure of scenes by learning to predict a scene representation from a single image. While these methods are typically trained end-to-end, they struggle to generalize to real-world photographs. The approaches most pertinent for generating 3D photographs of autochromes include those that leverage depth estimation followed by inpainting of occluded regions [24, 30, 11, 33, 20, 2]. 3D Photo [24] estimates monocular depth maps and employs layered depth images (LDIs) [22], where context-aware color and depth inpainting is used. We adopt this method [24] as it treats the scene as static, contrasting with approaches like 3D cinematography [13], which are inspired by space-time view synthesis techniques. This static representation is essential for autochromes since they are historical artifacts, and our objective is to avoid animating the scene’s objects to maintain authenticity.

3 METHODOLOGY

To extend the perception of autochromes from a 2D experience to an immersive 3D experience, we use a pipeline built on single-image monocular depth estimation, layered depth image generation, layered depth inpainting, and the production of a mesh (.glb file) for visualization in VR, thereby following the principle proposed by Kopf et al. [11]. In the following, we provide more details regarding the involved processing steps.

3.1 Depth estimation

Monocular or single-image depth estimation aims to extract the scene’s structure from a single image, relying on high-level monocular depth cues like occlusion boundaries and perspective, in contrast to settings where raw depth information is available from depth sensors or multi-view data with geometric constraints. We used MergeNet [16] along with the zero-shot depth estimator MiDaS [3] to estimate the depth of the 2D mono-autochromes and some of the examples are shown in Fig. 2.

3.2 Generation of layered depth image (LDI)

A layered depth image (LDI) [22] is a flexible 4-connected image that can store multiple pixels at each position, with each pixel con-



Figure 3: Example stereo view of 3D autochromes from a particular camera perspective

taining a color and depth value, and it explicitly represents local connectivity by storing pointers to at most one neighbor in each cardinal direction, allowing for connectivity in smooth regions while excluding neighbors across depth discontinuities. As proposed by Shih et. al. [24], we use the estimated depth to create a LDI. Given an input RGB-D image (depth estimated using single-image depth estimation as above), the method initializes a trivial LDI and detects major depth discontinuities, grouping them into connected depth edges that serve as the foundation for the inpainting process.



Figure 4: Example scenes used for conducting the user study.

3.3 Depth-aware inpainting

The inpainting algorithm [24] processes one depth edge at a time to synthesize new color and depth content in adjacent occluded regions, starting by disconnecting LDI pixels across the discontinuity, creating silhouette pixels. It generates a synthesis region using an iterative flood-fill algorithm, expanding from silhouette pixels to cover the occluded area while ensuring that the synthesis region remains strictly within this area and does not backtrack across silhouettes. Additionally, a context region is defined that only includes LDI pixels directly connected to the synthesis region, allowing the inpainting networks to focus solely on relevant surrounding content while addressing misalignments due to depth estimation by dilating the synthesis region near the depth edge.

3.4 Mesh generation

In the final stage of the pipeline, the depth-inpainted LDI is converted to a mesh [11]. The mesh is then simplified and converted to a Graphics Library Transmission Format (GLTF) to ensure web compatibility and facilitate faster transmission over the network. Due to the LDI's self-overlapping parts and complex topology, it cannot be mapped to a single contiguous texture image, necessitating the partitioning into flat charts that can be packed into an atlas image for texturing. The mesh simplification algorithm begins by converting the outline of each chart into a detailed 2D polygon and simplifying it using the Douglas-Peucker algorithm [4], ensuring shared boundaries between charts are simplified identically. Next, the chart interiors are triangulated by generating strips of vertical

polylines with evenly spaced interior vertices, followed by a fast plane-sweep algorithm for triangulation. Finally, the 2D triangulation is lifted to 3D by projecting each vertex along its corresponding ray according to its depth, resulting in a 3D triangle mesh.

3.5 Visualization in VR

In virtual reality (VR), two slightly different images are created for each eye to enhance depth perception and immersion. 3D autochromes fully utilize the six degrees of freedom available in exploration based on virtual reality technology, allowing for rich interaction. The process begins with rendering a 3D scene using a mesh generated from an LDI (as described in Sec. 3.4) through ThreeJS in a WebGL context. WebXR connects this rendered scene to a VR headset, producing stereo images (as shown in Fig. 3) tailored to the device's specifications — one for each eye. Additionally, the movement of the user's head directly translates to the virtual camera's motion, with constraints on viewing angles, further enhancing the immersive experience.

4 EVALUATION

We evaluated our technology by comparing the perception of 3D autochrome representations shown in VR to the traditional 2D perception of autochromes in their physical, analog form. For the latter, we utilized a backlight board, as shown in Fig. 6, that was taken during our study.

4.1 Implementation and dataset

The 3D autochromes (layered depth images and the corresponding meshes) were generated from selected digitized mono-autochrome images of the Harold Taylor collection [26] utilizing a machine equipped with an Nvidia RTX 3090 graphics card. The resulting WebXR application was developed using Three.js, and the user study was conducted with the Meta Quest 3 headset. Some of the scenes used for conducting the user study are presented in Fig. 4.

4.2 Study design

In a comparative AB within-subject design, each participant engaged with both conditions (A and B). In our study, condition A involves viewing analog autochrome images, while condition B entails experiencing 3D autochrome images through a VR headset. This approach facilitates a direct comparison between the two experiences while effectively controlling for individual differences. The study was conducted in a controlled laboratory environment (see Fig. 6 and Fig. 7) to minimize external influences and ensure the integrity of the results. It involved 12 unpaid and voluntary participants (8 male, 1 female, 1 diverse, and 2 who would not provide this information) between 21 and 57 years with an average age of 33.7 and standard deviation of 11.9 years. Their experience in

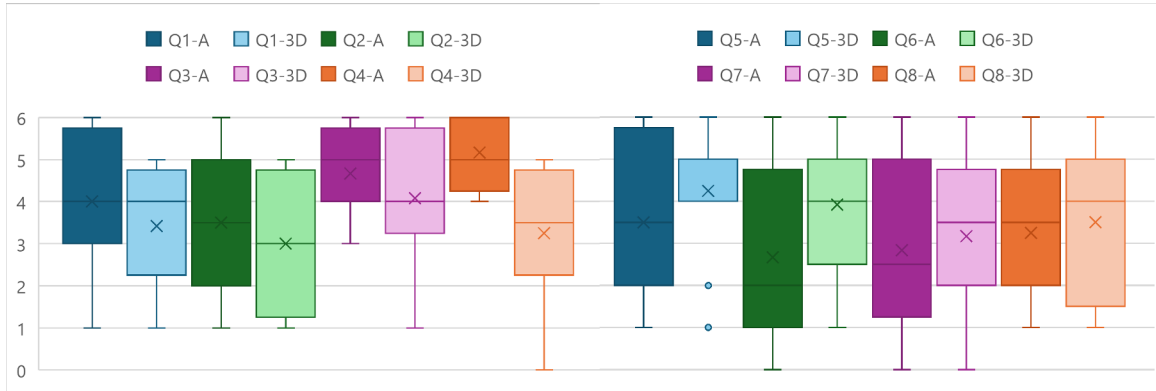


Figure 5: Boxplot visualization for Q1-Q8 for both conditions: Qx-A for analog/traditional experience and Qx-3D for our 3D VR experience.

general photography was asked using a four-point scale (1=no experience, 4=expert). With an average of 1.3 and standard deviation of 1.0, we classify our participant sample group as beginners but with high deviation. The user tests included two task conditions (*T-analog* and *T-3D*) to be accomplished by our participants using our software prototype. For *T-analog*, we asked them to explore a set of two physical autochromes with the backlight, whereas for *T-VR*, we asked to explore a set of five digitized 3D autochromes within our VR applications (randomized from a set of 17). We randomized the order that our participants performed the task conditions in, as well as the order and sub-set of autochromes within the tasks. For example, for our VR app, we had a total set of 17 3D autochromes prepared.

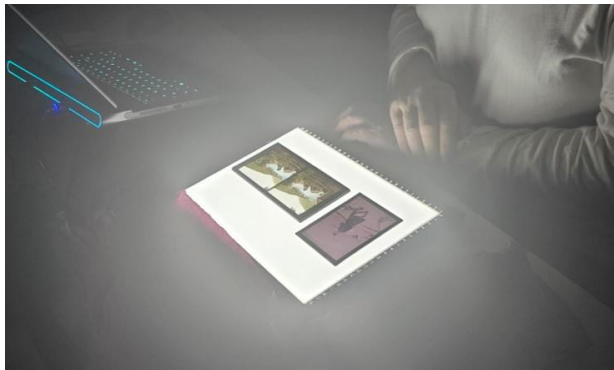


Figure 6: Example image taken during our user study where a participant explores physical autochrome photographs by means of a backlight board.

We asked our participants to fill out a questionnaire at four parts of our study: 1) at the beginning (introduction and demographic questions), 2&3) after each task, where the questionnaires incorporated the same questions, and 4) a post-study questionnaire at the end.

Questionnaires 2&3 incorporated the abbreviated version of the standardized AttrakDiff questionnaire [8], measuring the experience character by means of both pragmatic (PQ, e.g., usability) and hedonic qualities (HQ). In these questionnaires, we also asked eight custom questions on a 7-point semantic differential scale, where low numbers indicate a negatively connoted rating of the experience. Questions 1 to 5 captured aspects about the overall perception and aesthetic aspects of the autochrome experience, whereas questions 6 to 8 captured emotional aspects.

- Q1 How do you perceive the colours of the pictures? (very unnatural – very natural)
- Q2 How would you rate the image quality? (low – high)
- Q3 Did the pictures have a nostalgic or picturesque effect on you? (not at all – very much)
- Q4 How authentic did you find the pictures you looked at? (not authentic at all – very authentic)
- Q5 How detailed were you able to perceive the pictures? (not at all detailed – very detailed)
- Q6 How strongly did you feel immersed in the scene of the pictures? (not at all - very much)
- Q7 How strongly do you feel connected to the scene depicted in the picture? (not at all connected –very strongly connected)
- Q8 How strong were the emotions that the picture triggered in you? (no emotions – very strong emotions)

The post-study questionnaire incorporated further demographic questions and optional free-text questions about negative and positive aspects of the entire experience and suggestions for improvements, as well as the mandatory question *Q9: Which medium do you prefer for displaying an autochrome image? (analog or VR).*



Figure 7: Illustration of our study environment.

4.3 Study Results



Figure 8: Portfolio-presentation of the AttrakDiff values.

The results of the AttrakDiff questionnaire are visualized in the portfolio presentation in Fig. 8. It shows T-3D was assessed with higher hedonic than pragmatic quality whereas T-analog was assessed as rather balanced in this regard. The visualization places T-3D in the ‘self-oriented’ area and T-analog between ‘task-oriented’ and ‘desired’. Both confidence rectangles intersect with other areas of the continuum. Overall, our software used for T-3D obtained a higher average rating for HQ (4.2), but lower for PQ (3.3) than the traditional autochrome exploration (HQ 4.0, PQ 4.3).

The description of the word-pairs (Fig. 9) shows the mean value for each item of the AttrakDiff individually. Both experiences were rated positively, since only one item lies below the neutral mark of 3, which is ‘confusing–clearly structured’ for our 3D VR experience. The graph also shows that T-analog was perceived more pragmatic in all items except for ‘practical–impractical’ which is tied between both conditions. Regarding hedonic qualities, T-analog was rated more stylish and premium than our VR experience, which we can associate with the visual quality of our prototypical user interface implementation.

Figure 5 illustrates the results of Q1-Q8 as boxplots, split into the two test conditions, respectively. For Q1, Q2, Q3 and Q4 that relate to image quality and authenticity, the perception of the 3D autochrome mode is slightly inferior to the perception of the original 2D autochrome. However, for Q5, Q6, Q7 and Q8 that relate to the level of immersion, the perception of details and the triggering of connections and emotions, the potential of the 3D autochrome visualization gets apparent. Furthermore, our question asking which of the two experiences was preferred led to a result of 7 to 5 in favor of the analog exploration. We conducted Wilcoxon signed-rank tests [29] for the individual questions Q1-Q8 between the test conditions to analyze how significantly our experiences differ. With a threshold for statistical significance of 5%, no questions except for Q4 ($p=0.01$) show significant differences. We evaluated our participants’ free-text comments by identifying different themes and aspects, utilizing affinity diagrams as proposed by Preece, Rogers, and Sharp [19]. We could point out four aspects our participants

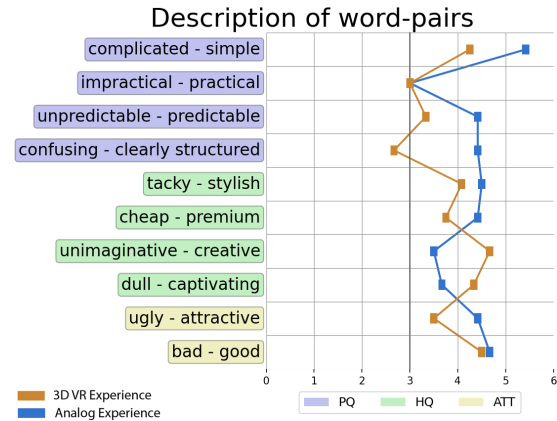


Figure 9: Word-pair visualization of the mean values of the AttrakDiff questionnaire.

mentioned. Firstly, the depth estimation we utilized was criticized, leading to ‘unnatural’ perceptions of the shown scene, specifically when humans were shown in the images. Secondly, our participants noted positively that the ability of having a large representation of the image did add to the overall experience and also to how detailed the pictures could be explored. The third aspect was a critique where the participants mentioned that the point of view of the 3D scene did look good only from one position, however, there was one participant extensively walking around and shifting the head position up and down, mentioning that some things would never have been explored without this feature. Lastly, our participants mentioned that such an VR application should make use of the interactivity that this medium can offer (‘Hardly any interaction ‘beyond’ watching’). Suggestions that were made are 1) highlighting ‘cool’ spots or spots ‘characteristic for autochrome photographs’ within the 3D imagery, 2) an option to view a 2D digital copy in the hands during 3D VR exploration, and 3) leveraging advantages of Augmented Reality or eXtended Reality technologies to offer exploring a 3D autochrome while also being able to interact with the physical one. Overall, our study results indicate that besides the authenticity (in favor of the analog experience), our participants found the 3D VR software for exploring autochrome photographs similarly pleasant (taken into account the AttrakDiff results) as the analog exploration. This underlines the potential that a 3D VR exploration can have in the future on digitally preserving the slowly decaying physical autochrome pictures. However, larger subject numbers and a more sophisticated user interface will be needed in order to get definitive insights.

5 LIMITATIONS

Despite the promising results of this study, several limitations must be acknowledged. Firstly, the depth estimation algorithms employed in our 3D reconstruction process could benefit from further refinement [33, 17]. Current algorithms may not accurately capture the complexities of the original images, leading to perceptions of unnatural depth in some scenes, particularly those featuring human subjects. Additionally, the visual quality of the autochromes in the virtual reality environment can be enhanced. Improvements in contrast and color fidelity are necessary to provide a more authentic representation of the original photographs.

Since autochromes are susceptible to degradation and defects including emulsion loss, dye bleeding, and glass cracking, utilizing artificial intelligence to remove these imperfections in 2D autochromes can enhance their quality and provide a more authentic viewing experience of 3D autochromes, thereby elevating the im-

mersive experience and better conveying their historical and cultural significance. Addressing these limitations through advanced depth estimation algorithms, restoration of defects, and improved image contrast will maximize the effectiveness of the 3D VR experience, ensuring a more engaging interaction with these valuable historical artifacts.

6 CONCLUSION

This study demonstrates the promising integration of deep learning technologies and 3D photography in reimagining the viewing experience of historical artifacts, specifically autochromes. By transforming digitized mono-autochromes into immersive three-dimensional representations, we can significantly enhance their cultural and historical significance, offering a richer engagement through virtual reality. User feedback highlights both the strengths and limitations of the current VR implementation. While many participants appreciated the immersive nature and the ability to explore images in greater detail, concerns were raised regarding the depth perception in certain scenes. These insights suggest that while the 3D VR medium can deliver a compelling alternative to traditional viewing methods, there remains room for improvement. To maximize the effectiveness of our approach, future work should focus on refining depth estimation, enhancing interactivity, advancing mesh simplification, increasing participant numbers, and utilizing a greater variety of analog autochromes for more robust studies. Addressing these areas will enrich the exploration experience of fragile historical artifacts like autochromes, ensuring their old is preserved and appreciated in a modern context.

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